

Network Flows (contd.)

* $\text{Flow} = \sum_{u: s \rightarrow u \in E} f(s \rightarrow u)$

* f is feasible if:

- $\forall u, v \in E$
 $0 \leq f(u \rightarrow v) \leq c(u \rightarrow v)$
- $\forall v \in V \setminus \{s, t\}$
 $\sum_{u: u \rightarrow v \in E} f(u \rightarrow v) = \sum_{w: v \rightarrow w \in E} f(v \rightarrow w)$

• Capacity of a cut (S, T) : $S \cup T = V$
 $\sum_{u \in S} \sum_{v \in T} c(u \rightarrow v)$

Theorem: If f is any feasible flow and (S, T) is any cut s.t. $s \in S$ and $t \in T$, then total flow $|f|$ is at most the capacity of the cut.

Proof: $|f| = \sum_{u: s \rightarrow u \in E} f(s \rightarrow u)$

$$= \sum_{u: s \rightarrow u \in E} f(s \rightarrow u) + \sum_{v \in V \setminus \{s, t\}} \overbrace{\left(- \sum_{u: u \rightarrow v \in E} f(u \rightarrow v) + \sum_{w: v \rightarrow w \in E} f(v \rightarrow w) \right)}^0$$

$$= \left(\sum_{v \in V \setminus \{t\}} \sum_w f(v \rightarrow w) \right) - \left(\sum_{v \in V \setminus \{s, t\}} \sum_u f(u \rightarrow v) \right)$$

$$\begin{aligned} & \left(\sum_{v \in S} \sum_{w \in S} f(v \rightarrow w) + \sum_{v \in S} \sum_{w \in T} f(v \rightarrow w) + \sum_{v \in T \setminus \{t\}} \sum_{w \in T} f(v \rightarrow w) \right) \\ & - \left(\sum_{v \in S} \sum_{u \in S} f(u \rightarrow v) + \sum_{v \in S} \sum_{u \in T} f(u \rightarrow v) + \sum_{v \in T \setminus \{t\}} \sum_{u \in T} f(u \rightarrow v) \right) \end{aligned}$$

$$= \sum_{v \in S} \sum_{w \in T} f(v \rightarrow w) - \sum_{v \in S} \sum_{u \in T} f(u \rightarrow v) \Rightarrow \text{THEOREM:}$$

Let f be any s-t flow and (A, B) be any s-t cut.
Then $f = f^{\text{out}}(A) - f^{\text{in}}(B)$

$$\leq \sum_{v \in S} \sum_{w \in T} f(v \rightarrow w)$$

$$\leq \sum_{v \in S} \sum_{w \in T} c(v \rightarrow w) = \text{capacity of } (S, T) \text{ cut}$$

$\therefore \underbrace{\text{Any feasible flow}}_{\text{maximise}} \leq \underbrace{\text{cut capacity}}_{\text{minimise}}$

Obs: A maximum flow can at most be the capacity of the minimum cut.

If min. cut $\leq$ max. flow, then min. cut capacity = max. flow

The algorithm terminates if there is no s-t path in the residual graph.



$(S) \leftarrow$ vertices reachable from source in the residual graph

↳ u is reachable from s in the residual graph
if $\exists$ a path from $s \rightarrow u$ w/ edges of > 0 residual capacity.

$$T = V \setminus S$$

- $\forall$ edges $u \rightarrow v \in E(G_{\text{original}})$
the capacity of these edges is saturated.
 $f(u \rightarrow v) = c(u \rightarrow v)$

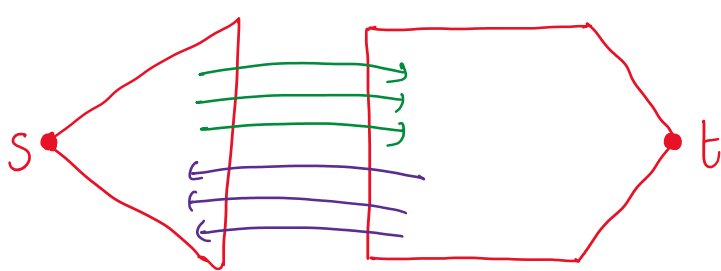
If not, i.e. if $f(u \rightarrow v) < c(u \rightarrow v)$,
then v could have been reached from source in the residual graph.

- $\forall u \in T, v \in S$ s.t. $u \rightarrow v \in E(G_{\text{original}})$,
the flow on these edges = 0.
If not, i.e. if $f(u \rightarrow v) > 0$ then in the residual graph
 $c(v \rightarrow u) = f(u \rightarrow v)$ and thus u could have been reachable.

Flow is at least $\sum_{u \in S} \sum_{v \in T} c(u \rightarrow v)$

(coupled with the fact that feasible flow $\leq$ capacity of any cut)

We get max flow and min cut are obtained by the algorithm.



$$\text{Net flow across the cut} = \left(\sum_{u \in S} \sum_{v \in T} f(u \rightarrow v) \right) - \left(\sum_{u \in T} \sum_{v \in S} f(u \rightarrow v) \right)$$

When algo. terminates: $\sum_{u \in S} \sum_{v \in T} c(u \rightarrow v) \quad \parallel \quad 0$

(If there are parallel edges, make them one edge with combined capacity.)

Obs: This works as long as we do not have 2-cycles in the underlying undirected graph.



This can be fixed by modifying the underlying graph

